

Exercise_04_SQL_Joins

DT MBENGA

DATA ANALYTICS

BRIGHTLEAN

13/09/2026

Q1

SQL Editor

Exercise 04 - Question-1 × Exercise 04- Question -2 Exercis

Run all (1000) ✓ 07:28 PM (23s) workspace. default Last edit was 48 minutes

```
1 | DROP TABLE IF EXISTS users;
2 | CREATE TABLE users (
3 |     user_id INT,
4 |     user_name STRING,
5 |     country STRING
6 | );
7 | INSERT INTO users VALUES
8 | (1, 'Nomvula', 'Johannesburg'),
9 | (2, 'David', 'Cape Town'),
10 | (3, 'Anele', 'Durban'),
11 | (4, 'Kabelo', 'Pretoria'),
12 | (5, 'Lerato', 'Port Elizabeth');
13 | DROP TABLE IF EXISTS subscriptions;
```

Add parameter

← Results 7 of 7 → Table +

	1 ² ₃ user_id	A ^B _C user_name	1 ² ₃ subscription_id	📅 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

Q2

SQL Editor

Exercise 04- Question -2 × Exercise 04- Question-3 Exercise 04- Qu

Run all (1000) ✓ 07:41 PM (21s) workspace default Last edit was 35 minutes ago

```
9      (11, 'Standard', 129),
10     (12, 'Premium', 199),
11     (13, 'Family', 249),
12     (14, 'Mobile', 59);
13  /* Q2: Join subscriptions and plans */
14  SELECT
15      s.subscription_id,
16      s.user_id,
17      p.plan_name,
18      p.monthly_price
19  FROM subscriptions s
20  INNER JOIN plans p
21      ON s.plan_id = p.plan_id;
```

Add parameter

← Results 4 of 4 → Table +

>	subscription_id	user_id	plan_name	monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

Q3

```
23 (903, 1, 702, 60),
24 (904, 7, 701, 20),
25 (905, 3, 705, 90);
26 /* Q3: Join viewing sessions and shows */
27 SELECT
28     v.session_id,
29     v.user_id,
30     s.show_title,
31     s.genre,
32     v.watch_minutes
33 FROM viewing_sessions v
34 INNER JOIN shows s
35     ON v.show_id = s.show_id;
```

Add parameter

← Results 7 of 7 →

Table ▾

+

	1 ² ₃ session_id	1 ² ₃ user_id	A ^B _C show_title	A ^B _C genre	1 ² ₃ watch_minutes
1	901	1	Comedy Hour	Comedy	45
2	902	2	Tech Tales	Documentary	30
3	903	1	Crime Time	Drama	60
4	904	7	Comedy Hour	Comedy	20

Q4

```
1  /* Q4: Join users and viewing sessions */
2
3  SELECT
4      u.user_name,
5      u.country,
6      v.session_id,
7      v.show_id,
8      v.watch_minutes
9  FROM users u
10 INNER JOIN viewing_sessions v
11      ON u.user_id = v.user_id;
```

Add parameter

Table  

	A_C^B user_name	A_C^B country	1_3^2 session_id	1_3^2 show_id	1_3^2 watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	901	701	45

Q5

```
1  /* Q5: Join users, subscriptions, and plans */
2
3  SELECT
4      u.user_name,
5      u.country,
6      p.plan_name,
7      p.monthly_price,
8      s.start_date
9  FROM users u
10 INNER JOIN subscriptions s
11     ON u.user_id = s.user_id
12 INNER JOIN plans p
13     ON s.plan_id = p.plan_id;
```

Add parameter

Table  

	 user_name	 country	 plan_name	 monthly_price	 start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10
2	David	Cape Town	Standard	129	2026-02-01
3	Anele	Durban	Family	249	2026-04-05
4	Nomvula	Johannesburg	Basic	79	2026-01-15

Q6

Run all (1000) ✓ 07:52 PM (1s) workspace.default Last edit was 32 minutes ago

1

2

3

4

5

6

7

8

9

10

/* Q6: Show all users and their subscriptions */

SELECT

u.user_id,

u.user_name,

s.subscription_id,

s.start_date

FROM users u

LEFT JOIN subscriptions s

ON u.user_id = s.user_id;

Add parameter

Table +

>

	1 ² ₃ user_id	Δ ^B _C user_name	1 ² ₃ subscription_id	📅 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15

Q7

▶ Run all (1000) ✓ 07:55 PM (1s) workspace.default Last edit was 30 minutes ago

```
1  /* Q7: Show all plans and their subscriptions */
2
3  SELECT
4      p.plan_id,
5      p.plan_name,
6      s.subscription_id,
7      s.user_id
8  FROM plans p
9  LEFT JOIN subscriptions s
10     ON p.plan_id = s.plan_id;
```

Add parameter

Table +

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Q8

Run all (1000) ✓ 07:56 PM (1s) workspace. default Last edit was 30 minutes ago

1

2

3

4

5

6

7

8

9

10

/* Q8: Show all shows and their viewing sessions */

SELECT

s.show_id,

s.show_title,

v.session_id,

v.watch_minutes

FROM shows s

LEFT JOIN viewing_sessions v

ON s.show_id = v.show_id;

Add parameter

Table +

>

	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

Q9

```
1  /* Q9: Show all viewing sessions and matching users */
2
3  SELECT
4      v.session_id,
5      v.show_id,
6      v.watch_minutes,
7      u.user_id,
8      u.user_name
9  FROM viewing_sessions v
10 LEFT JOIN users u
11      ON v.user_id = u.user_id;
```

Add parameter

Table  

	^{1 2} ₃ session_id	^{1 2} ₃ show_id	^{1 2} ₃ watch_minutes	^{1 2} ₃ user_id	^{A B} _C user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele

Q10

```
2  /* Q10: Show all users and their plan information if available */
3
4  SELECT
5      u.user_name,
6      u.country,
7      p.plan_name,
8      p.monthly_price
9  FROM users u
10 LEFT JOIN subscriptions s
11      ON u.user_id = s.user_id
12 LEFT JOIN plans p
13      ON s.plan_id = p.plan_id;
```

Add parameter

Table  

	^A _C user_name	^A _C country	^A _C plan_name	¹ ₃ monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Port Elizabeth	null	null
6	Nomvula	Johannesburg	Basic	79

Q11

```
2
3  /* Q11: Show all users and subscriptions, including unmatched records */
4
5  SELECT
6      COALESCE(u.user_id, s.user_id) AS user_id,
7      u.user_name,
8      s.subscription_id,
9      s.start_date
10 FROM users u
11 FULL OUTER JOIN subscriptions s
12     ON u.user_id = s.user_id;
```

Add parameter

Table  

	 user_id	 user_name	 subscription_id	 start_date
1	5	Lerato	 null	 null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	 null	 null
6	1	Nomvula	501	2026-01-15
7	6	 null	504	2026-03-20

Q12

```
1
2
3  /* Q12: Show all plans and subscriptions, including unmatched records */
4
5  SELECT
6      COALESCE(p.plan_id, s.plan_id) AS plan_id,
7      p.plan_name,
8      s.subscription_id,
9      s.user_id
10 FROM plans p
11 FULL OUTER JOIN subscriptions s
12     ON p.plan_id = s.plan_id;
```

Add parameter

Table  

	¹ ₃ plan_id	^A _C plan_name	¹ ₃ subscription_id	¹ ₃ user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

Q13

```
2
3  /* Q13: Show all shows and viewing sessions, including unmatched records */
4
5  SELECT
6      COALESCE(s.show_id, v.show_id) AS show_id,
7      s.show_title,
8      v.session_id,
9      v.watch_minutes
10 FROM shows s
11 FULL OUTER JOIN viewing_sessions v
12     ON s.show_id = v.show_id;
```

Add parameter

Table ▾ +

	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	705	null	905	90

Q14

```
1
2  /* Q14: Show all users and viewing sessions, including unmatched records */
3
4  SELECT
5      COALESCE(u.user_id, v.user_id) AS user_id,
6      u.user_name,
7      v.session_id,
8      v.show_id,
9      v.watch_minutes
10 FROM users u
11 FULL OUTER JOIN viewing_sessions v
12     ON u.user_id = v.user_id;
```

add parameter

table ▾ +

	¹² ₃ user_id	^A _C user_name	¹² ₃ session_id	¹² ₃ show_id	¹² ₃ watch_minutes
1	1	Nomvula	901	701	45
2	1	Nomvula	903	702	60
3	3	Anele	905	705	90
4	2	David	902	703	30
5	7	null	904	701	20
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

Q15

```
/* Q15: Show all users, subscriptions, and plans using FULL OUTER JOIN */
SELECT
    COALESCE(u.user_id, s.user_id) AS user_id,
    u.user_name,
    s.subscription_id,
    COALESCE(s.plan_id, p.plan_id) AS plan_id,
    p.plan_name
FROM users u
FULL OUTER JOIN subscriptions s
    ON u.user_id = s.user_id
FULL OUTER JOIN plans p
    ON s.plan_id = p.plan_id;
```

parameter

e v +

¹ ₂ ³ user_id	^A _B ^C user_name	¹ ₂ ³ subscription_id	¹ ₂ ³ plan_id	^A _B ^C plan_name
1	Nomvula	503	12	Premium
1	Nomvula	501	10	Basic
3	Anele	505	13	Family
5	Lerato	null	null	null
4	Kabelo	null	null	null
2	David	502	11	Standard
6	null	504	11	Standard
null	null	null	14	Mobile

Q16

```
1
2  /* Q16: Find users who do not have a subscription */
3
4  SELECT
5      u.user_id,
6      u.user_name
7  FROM users u
8  LEFT JOIN subscriptions s
9      ON u.user_id = s.user_id
10 WHERE s.subscription_id IS NULL;
```

Add parameter

Table



	1^2_3 user_id	A^B_C user_name
1	4	Kabelo
2	5	Lerato

Q17

```
1
2  /* Q17: Find subscriptions referencing non-existent users */
3
4  SELECT
5      s.subscription_id,
6      s.user_id,
7      s.plan_id
8  FROM subscriptions s
9  LEFT JOIN users u
10     ON s.user_id = u.user_id
11  WHERE u.user_id IS NULL;
```

Add parameter

Table  

	¹ ₂ subscription_id	¹ ₂ user_id	¹ ₂ plan_id
1	504	6	11

Q18

```
1
2
3  /* Q18: Find shows that have never been watched */
4
5  SELECT
6      s.show_id,
7      s.show_title
8  FROM shows s
9  LEFT JOIN viewing_sessions v
10     ON s.show_id = v.show_id
11     WHERE v.session_id IS NULL;
```

Add parameter

Table  

	1^2_3 show_id	A^B_C show_title
1	704	Cooking Lab
2	706	Wild Earth

Q19

1

2

3

4

5

6

7

8

9

10

11

```
/* Q19: Find viewing sessions referencing non-existent shows */

SELECT
    v.session_id,
    v.user_id,
    v.show_id
FROM viewing_sessions v
LEFT JOIN shows s
    ON v.show_id = s.show_id
WHERE s.show_id IS NULL;
```

Add parameter

Table ▾ +

>	¹ ₃ session_id	¹ ₃ user_id	¹ ₃ show_id
1	905	3	705

Q20

1

2

3

4

5

6

7

8

9

10

/* Q20: Find plans with no subscribers */

SELECT

p.plan_id,

p.plan_name

FROM plans p

LEFT JOIN subscriptions s

ON p.plan_id = s.plan_id

WHERE s.subscription_id IS NULL;

Add parameter

Table

+

>

1²₃ plan_id

A^B_C plan_name

1

14

Mobile